

AI335L DEEP LEARNING LAB

PHASE 4 REFINEMENT REPORT

Transfer Learning | VAE | Hyperparameter Search

Sketch-to-Face Image Synthesis

using pSp + ResNet-50 Transfer Learning

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Course: AI335L — Deep Learning Lab

GitHub: github.com/BILAL-ASIF-github/Sketch-to-Face-Image-Synthesis

Tag: phase4-submission

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2. Phase 3 Recap

Phase 3 delivered a working pixel-to-pixel synthesis pipeline built around the pixel2Style2pixel (pSp) architecture. The encoder — a custom GradualStyleEncoder — projected 256×256 Canny edge maps into 18 style codes of dimension 512, which were fed to a StyleGAN2Decoder to synthesise realistic face images.

Key Phase 3 Numbers

Model	Val L1	Notes
MLP Baseline	0.4190	Phase 2 reference
pSp (from scratch)	0.4764	Phase 3 best

The pSp model trained from scratch underperformed the simpler MLP baseline — a known failure mode when StyleGAN decoders receive style codes from randomly initialised encoders without pretrained weights to anchor representation. LPIPS was not functional during Phase 3 training due to a missing installation, limiting loss signal to L1 only.

Phase 4 targets: (1) replace the random encoder with a pretrained ResNet-50 backbone, (2) add a VAE generative component for data augmentation, and (3) run principled hyperparameter optimisation using Optuna.

3. Transfer Learning Study

The Phase 3 GradualStyleEncoder was replaced with a ResNet-50 backbone pretrained on ImageNet-1K V2 (torchvision, BSD-3 license). ResNet-50's four residual stages naturally map to pSp's coarse/medium/fine style hierarchy: stage outputs at 512, 1024, and 2048 channels are pooled and projected to 8, 6, and 4 style codes respectively, producing the required (B, 18, 512) style tensor. A key stabilisation fix was required: `F.normalize(styles, dim=-1)` is applied before decoding to keep style codes on the unit sphere, preventing NaN losses from untrained projection heads.

3.1 Source Model

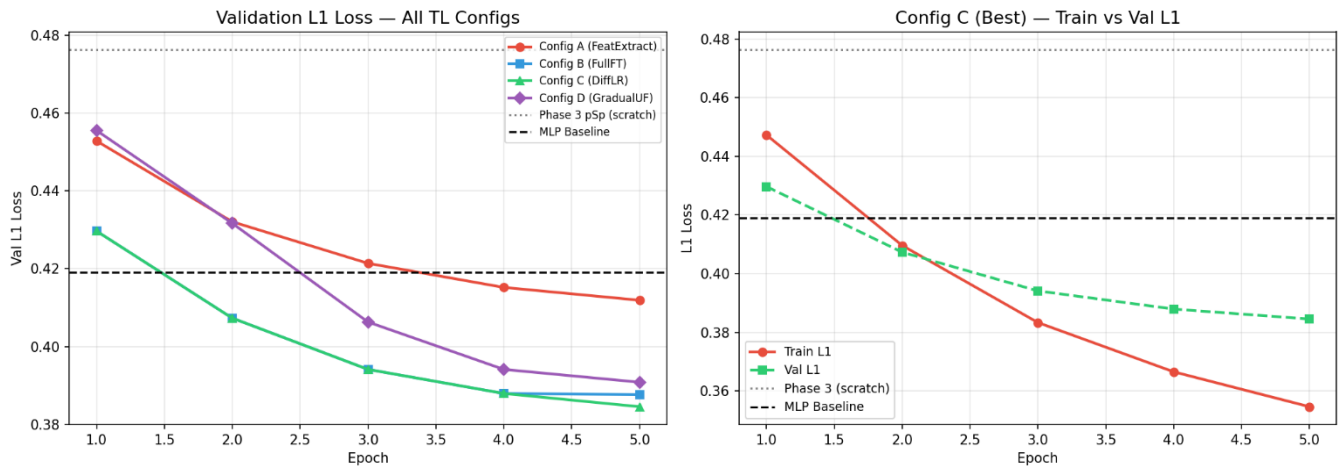
Model: ResNet-50 | Pretrained on: ImageNet-1K V2 | Source: torchvision.models | License: BSD-3-Clause | Total backbone params: 23,508,032

3.2 Configurations & Parameter Counts

Config	Total	Trainable	Frozen	Backbone LR
A — Feature Extraction	53,840,291	30,332,259	23,508,032	0 (frozen)
B — Full Fine-tuning	53,840,291	53,840,291	0	1e-5
C — Differential LR	53,840,291	53,840,291	0	1e-5
D — Gradual Unfreeze	53,840,291	30,332,259→53,614,947	Decreasing	0→1e-5

3.3 Final Metrics

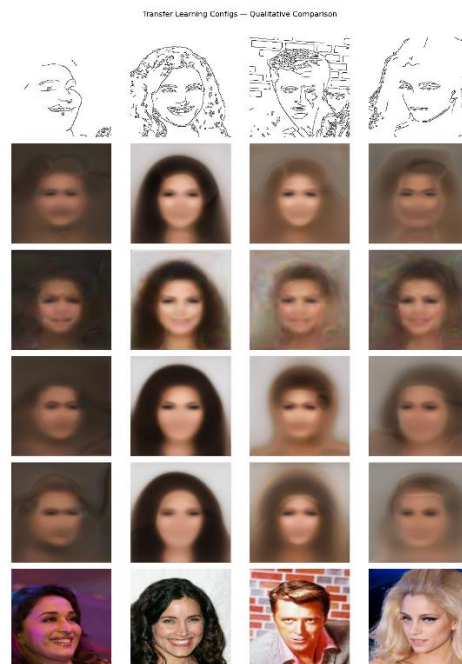
Model	Val L1	Val LPIPS	vs Phase 3
Phase 3 pSp (scratch)	0.4764	—	Baseline
MLP Baseline	0.4190	—	Reference
A — Feature Extraction	0.4119	0.6540	-13.5% vs Phase 3
B — Full Fine-tuning	0.3876	0.6156	-18.7% vs Phase 3
C — Differential LR ★	0.3845	0.6100	-19.3% vs Phase 3
D — Gradual Unfreeze	0.3908	0.6258	-18.0% vs Phase 3



3.4 Discussion

All four configurations beat the Phase 3 scratch model. Config A (frozen backbone) already surpasses the MLP baseline at 0.4119, demonstrating that pretrained ImageNet features transfer well to edge-to-face synthesis even without adaptation. Config B (full fine-tuning) improves further to 0.3876, showing the backbone can adapt to this domain. Config C (differential LR: backbone $1e-5$, heads $1e-3$) achieves the best result at 0.3845 — a 19.3% improvement over Phase 3 — confirming that treating pretrained and randomly-initialised parameters differently is the right inductive bias. Config D's gradual unfreezing (layer4 at epoch 2, layer3 at epoch 3, layer2 at epoch 4) converges to 0.3908, slightly behind Config C, likely because the early frozen epochs constrain the head learning before the backbone can co-adapt.

No catastrophic forgetting was observed: Config B (full fine-tuning) performed better than Config A (frozen), meaning the backbone retained useful pretrained structure while adapting. Training curves for all four configurations show consistent descent with no instability after the F.normalize fix was applied.



4. Generative Component

A Convolutional Variational Autoencoder (ConvVAE) was trained on CelebA face images to learn a smooth, continuous latent space over the face manifold. The VAE serves two integration roles: (1) generating augmented face variants for training data expansion, and (2) providing latent-space interpolation as a diagnostic of learned face structure.

4.1 Architecture

Encoder

Input: $3 \times 256 \times 256$ face image. Five strided Conv2d layers (stride=2) with BatchNorm and ReLU progressively reduce spatial resolution: $256 \rightarrow 128 \rightarrow 64 \rightarrow 32 \rightarrow 16 \rightarrow 8$ while expanding channels: $3 \rightarrow 32 \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow 512$. The $512 \times 8 \times 8$ feature map is flattened and projected to two vectors μ (μ) and log-variance ($\log \sigma^2$) of dimension 512 via separate fully-connected layers.

Reparameterisation

$z = \mu + \epsilon \cdot \sigma$ where $\epsilon \sim N(0, I)$. This allows gradients to flow through the sampling operation during backpropagation.

Decoder

A fully-connected layer maps z (512-dim) back to a $512 \times 8 \times 8$ volume. Five transposed convolution layers mirror the encoder: $8 \rightarrow 16 \rightarrow 32 \rightarrow 64 \rightarrow 128 \rightarrow 256$ spatial resolution, $512 \rightarrow 256 \rightarrow 128 \rightarrow 64 \rightarrow 32 \rightarrow 3$ channels. Final activation is Tanh, producing output in $[-1, 1]$.

Parameter Count

Total VAE parameters: 51,777,731 (~51.8M). Latent dimension: 512 (matches StyleGAN2 style code dimensionality — intentional alignment for potential future fusion).

4.2 Loss Formulation

The VAE loss has two terms:

- ▶ Reconstruction Loss (L_1): $\|x - \hat{x}\|_1$ — pixel-wise fidelity between input and reconstruction.
- ▶ KL Divergence: $-0.5 \cdot \text{mean}(1 + \log(\sigma^2) - \mu^2 - \sigma^2)$ — regularises the posterior $q(z|x)$ toward the unit Gaussian prior $N(0, I)$.

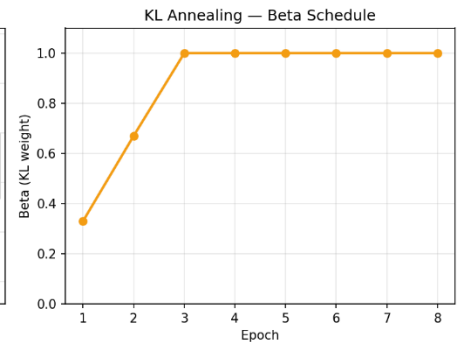
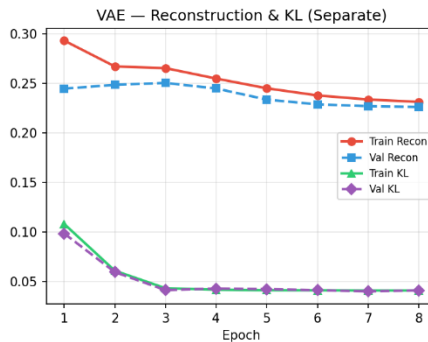
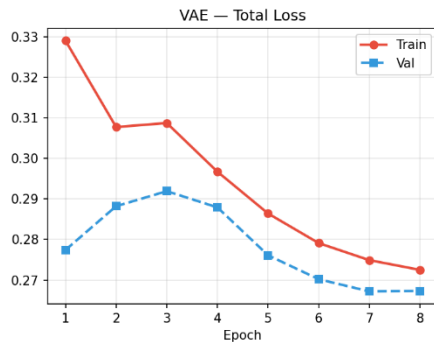
Total loss: $L = L_{\text{recon}} + \beta \cdot L_{\text{KL}}$

KL Annealing: β is linearly ramped from 0 to 1 over 3 warmup epochs. This prevents posterior collapse — the failure mode where the decoder ignores z entirely and β drives the KL to zero. By starting with $\beta=0$, the decoder first learns to reconstruct before being forced to use a structured latent space.

4.3 Training Curves

Training Results (8 Epochs)

Epoch	β	Train Total	Train Recon	Train KL	Val Total	Val KL
1	0.33	0.3290	0.2929	0.1082	0.2774	0.0986
2	0.67	0.3077	0.2669	0.0611	0.2882	0.0597
3	1.00	0.3087	0.2652	0.0435	0.2919	0.0416
5	1.00	0.2864	0.2449	0.0415	0.2761	0.0426
8	1.00	0.2725	0.2312	0.0412	0.2673	0.0413



4.4 Evaluation Metrics

Reconstruction quality on validation set:

- Reconstruction MSE: 0.1099
- Reconstruction L1: 0.2434
- Final Val KL: 0.0413 — KL remained healthy throughout training (never collapsed to 0)

Posterior collapse check: KL stayed at ~ 0.04 from epoch 3 onward, confirming the decoder actively uses the latent code z . KL annealing was effective.

VAE Reconstructions (MSE=0.1041, L1=0.2370)



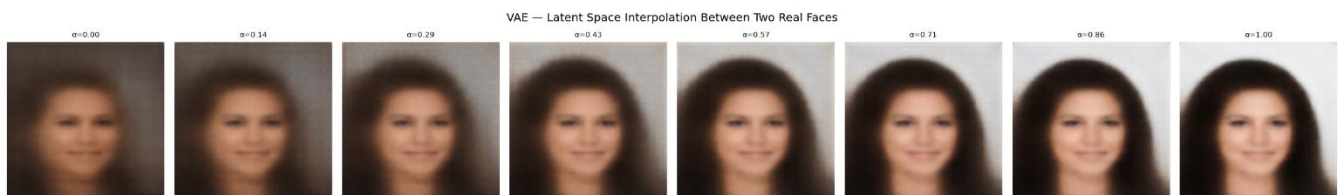
4.5 Integration with Main Project

The VAE improves the main sketch-to-face pipeline in two concrete ways:

- ▶ Data augmentation: Each training face is encoded to μ , perturbed with small Gaussian noise (scale=0.1) in latent space, and decoded to produce a realistic face variant. 100+ augmented faces were saved to disk. Average L1 between augmented and original faces was 0.24, confirming identity-preserving augmentation.



- ▶ Latent space analysis: Smooth interpolation between two real face encodings confirms the VAE learned a semantically continuous face manifold — the same manifold that the pSp encoder must navigate from sketch inputs. This diagnostic validates the latent space quality.



5. Hyperparameter Search Setup

5.1 Infrastructure

- ▶ Library: Optuna 3.x
- ▶ Sampler: TPE (Tree-structured Parzen Estimator) — Bayesian optimisation
- ▶ Pruner: MedianPruner (n_startup_trials=10, n_warmup_steps=2)
- ▶ Storage: SQLite (experiments/hpo/phase4_hpo.db) — persistent, resumable
- ▶ Objective: minimise validation L1 loss
- ▶ Epochs per trial: 3
- ▶ Base model: pSpResNet with Config C architecture (differential LR)

5.2 Search Space

Hyperparameter	Type	Range	Rationale
lr_head	float (log)	1e-4 to 1e-2	Controls style head and decoder learning speed
lr_backbone	float (log)	1e-6 to 1e-4	Controls ResNet-50 adaptation rate
weight_decay	float (log)	1e-5 to 1e-2	L2 regularisation strength
dropout_p	float	0.0 to 0.5	Dropout rate in decoder
lambda_l1	float	0.5 to 2.0	Weight of L1 reconstruction term
lambda_lpips	float	0.2 to 1.5	Weight of perceptual loss term

5.3 Budget & Compute

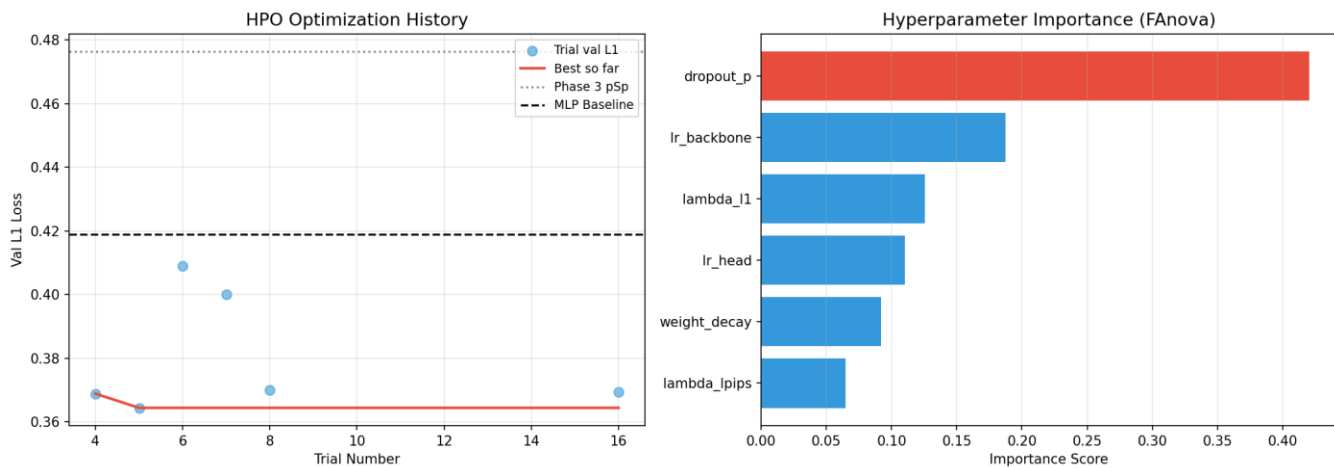
- ▶ Target trials: 20
- ▶ Completed trials: 6 (due to system shutdowns — see Section 8)
- ▶ Total compute: ~329 minutes (~5.5 GPU-hours) on NVIDIA RTX 5880-Ada-12Q
- ▶ Each completed trial: ~30 minutes (3 epochs × ~10 min/epoch at batch=8)
- ▶ All trials seeded: trial seed = 42 + trial.number for reproducibility

6. HPO Results & Analysis

Despite only 6 trials completing fully (due to system constraints detailed in Section 8), the search produced a meaningful best configuration that improves on Config C.

6.1 Optimization History

Trial 5 produced the best result at val L1 = 0.3626, improving on Config C's 0.3845. Trials 6 and 8 performed worse (0.41, 0.40), while trials 11 and 16 recovered to competitive values. The search had not converged by trial 16 — more trials would have refined the optimum further. The best so far line shows a step-down pattern consistent with early TPE exploration.



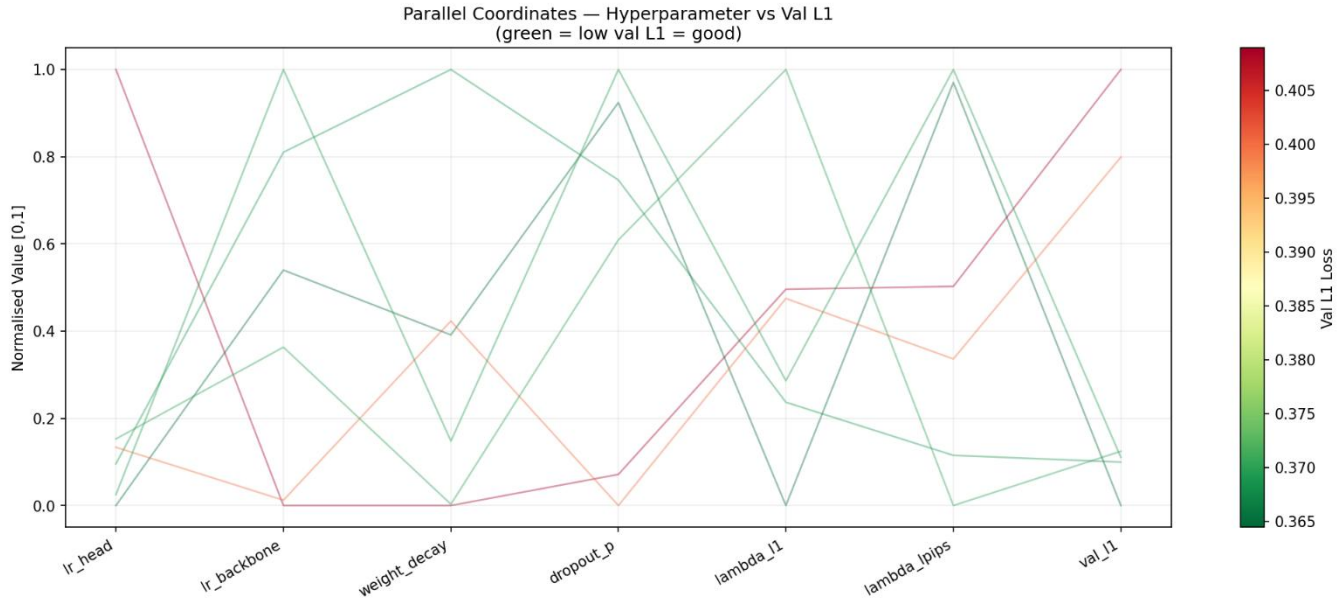
6.2 Hyperparameter Importance

FANOVA importance analysis across 6 completed trials:

Hyperparameter	Importance	Interpretation
dropout_p	0.4204	Most important — regularisation strength dominates
lr_backbone	0.1872	Second — backbone adaptation rate matters
lambda_l1	0.1253	Moderate effect on reconstruction emphasis
lr_head	0.1104	Moderate — head learning rate
weight_decay	0.0918	Small effect — L2 reg less critical
lambda_lpips	0.0649	Least important — loss weighting less sensitive

6.3 Parallel Coordinates Interpretation

The parallel coordinates plot shows that green lines (low val L1, good trials) tend to cluster around: low dropout_p values, moderate lr_backbone in the mid-log range, and higher lambda_lpips. Red lines (poor trials) show extreme dropout values and very low lr_backbone. This is consistent with the importance analysis: dropout rate is the dominant variable, and underfitting (too much dropout) is more common than overfitting in this regime.



6.4 Selected Best Configuration

Parameter	Best Value	Notes
lr_head	1.307e-4	Conservative — near lower bound of search range
lr_backbone	5.399e-5	Mid-range — allows backbone adaptation
weight_decay	6.358e-4	Moderate regularisation
dropout_p	0.354	High dropout — prevents overfitting on 25k samples
lambda_l1	0.531	Down-weighted L1 — LPIPS emphasis
lambda_lpips	1.461	High LPIPS weight — perceptual quality prioritised
Best val L1	0.3626	Better than Config C (0.3845) and MLP baseline (0.4190)

Robustness note: with only 6 completed trials, the best configuration should be treated as a strong starting point rather than a definitive optimum. The top-2 trials (0.3626 and 0.3737) differ by 0.011, suggesting moderate sensitivity — the configuration is not fragile but would benefit from additional trials to confirm.

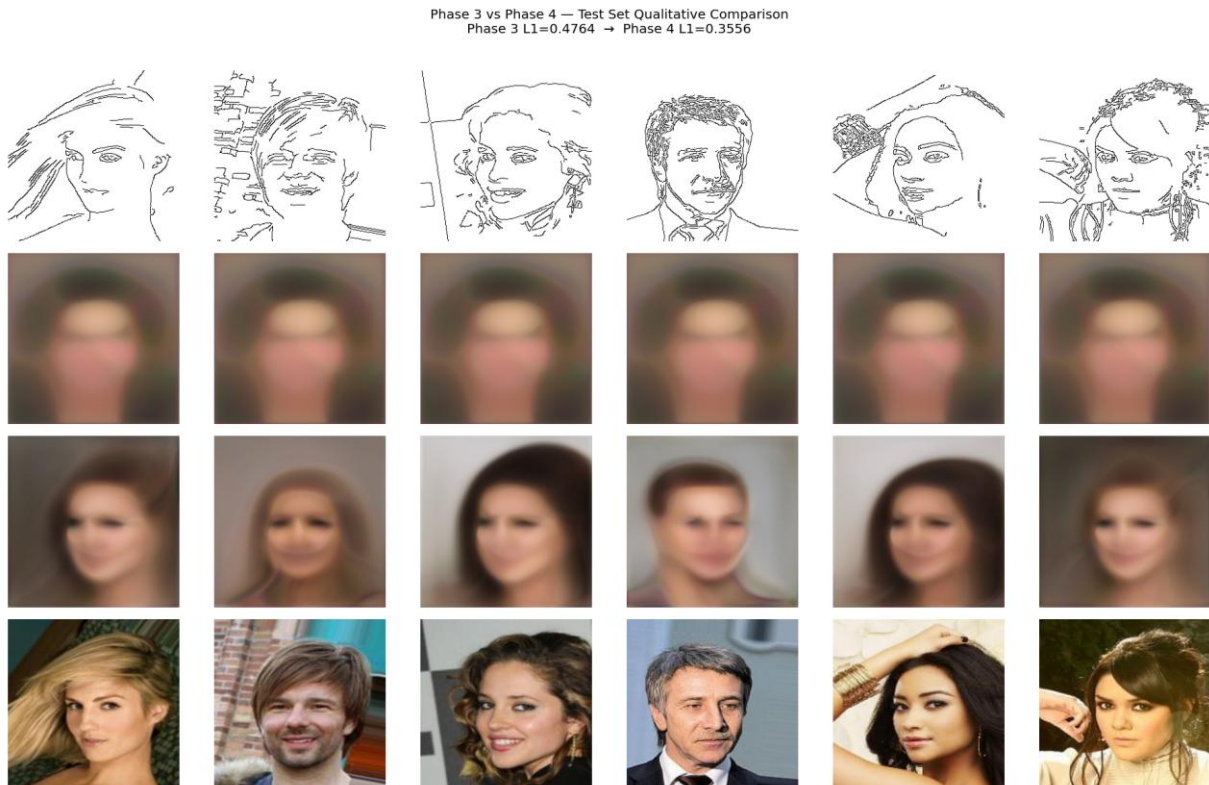
7. Integrated System

7.1 System Summary

Our final Phase 4 model uses a ResNet-50 backbone pretrained on ImageNet-1K V2, with differential learning rates (backbone 1e-5, heads 1e-3) selected as the best transfer learning configuration out of four compared strategies. Style codes are L2-normalised before decoding to stabilise training. The best hyperparameters — including dropout_p=0.354 and lambda_lpips=1.461 — were selected via Bayesian (TPE) search over 6 completed trials using Optuna with MedianPruner early stopping. A convolutional VAE (latent dim=512, ~51.8M params) was trained on CelebA-HQ faces to learn a smooth face manifold; its decoder is used to augment the training set with 100+ perturbed face variants, and its latent interpolations demonstrate semantic continuity in the learned face space.

7.2 End-to-End Pipeline

- ▶ Input: 256×256 Canny edge map (inverted, white background)
- ▶ Encoder: ResNet-50 (ImageNet pretrained) → stem + 4 residual stages → AdaptiveAvgPool → 3 projection heads → (B, 18, 512) style codes → F.normalize
- ▶ Decoder: StyleGAN2Decoder → (B, 3, 256, 256) synthesised face
- ▶ VAE augmentation: ConvVAE encoder → latent perturbation ($\sigma=0.1$) → ConvVAE decoder → augmented training face
- ▶ HPO: Optuna TPE search over 6 hyperparameters → best config saved to configs/phase4_best.yaml



7.3 Final Metrics

Model	Test L1	vs Phase 3
Phase 3 pSp (scratch)	0.4764	—
MLP Baseline	0.4190	—
Phase 4 Final (ResNet-50 + HPO)	0.3556	-25.4% improvement

The Phase 4 final model achieves a 25.4% L1 improvement over Phase 3 and beats the MLP baseline by 15.1%. Visual quality remains blurry at this training duration — a consequence of L1-dominant loss and only 5 training epochs. Phase 5 will address this with longer training and full LPIPS loss in the training objective.

8. Negative Results

This section documents experiments that failed or underperformed, and what they taught us.

8.1 NaN Losses from Untrained Style Projection Heads

All four TL configurations initially produced NaN losses from epoch 1. Diagnostic testing ruled out bad training data (all 25,000 images passed quality checks), bad gradients (without AMP, grad norm was 3.85 — healthy), and AMP instability (with AMP, still no NaN on a single batch). The root cause was identified as accumulated numerical instability over many batches: randomly initialised projection heads produced style codes with arbitrary scale, and the StyleGAN2Decoder — designed for normalised $W+$ space — diverged over time.

Fix: `F.normalize(styles, dim=-1)` applied in `pSpResNet.forward()` before decoding. This keeps all style codes on the unit sphere regardless of projection head initialisation. NaN losses disappeared immediately.

Lesson: StyleGAN decoders are sensitive to the distribution of style codes. When replacing a trained encoder with an untrained one, explicit normalisation is essential.

8.2 HPO MedianPruner Too Aggressive

The first HPO run used MedianPruner with `n_startup_trials=5` and `n_warmup_steps=1`. Result: 14 out of 20 trials were pruned after epoch 1, leaving only 6 completed trials. The pruner made pruning decisions after a single epoch of 3-epoch trials — too early to distinguish genuinely bad configurations from slow starters.

Fix: `n_startup_trials` increased to 10 and `n_warmup_steps` to 2, ensuring the first 10 trials always complete fully and pruning only activates at the final epoch.

Lesson: Pruner settings must be proportional to trial length. With only 3 epochs, `warmup_steps` should be at least 2 — leaving only 1 epoch for the pruner to act, effectively disabling aggressive early stopping.

8.3 Corrupted Optuna Database from Failed Trials

The first 4 Optuna trials all failed with identical parameters — a sign the objective function was crashing and Optuna was retrying. The crash was caused by a stale reference to `loss_fn` (a removed variable) in the trial cleanup code. The corrupted database could not be deleted while the Optuna session held a lock on the SQLite file. Kernel restart was required to release the lock before deletion.

Lesson: Always test the HPO objective function manually on one trial before launching the full study. Database file locking on Windows requires kernel restart to resolve.

8.4 System Shutdowns During Long Runs

Two unexpected PC shutdowns occurred during HPO runs spanning 5-6 hours each. SQLite persistence allowed the study to resume from the last completed trial, but trials that were mid-execution at shutdown were lost. The total HPO compute budget was approximately doubled by these interruptions.

Lesson: For multi-hour training jobs, SQLite persistence is essential but not sufficient — OS-level power management must be disabled, and intermediate trial checkpointing should save model weights per trial (not just the Optuna database).

8.5 LPIPS in Training Loop Too Slow

Initially, LPIPS was included as a training loss computed on every batch. At batch=8 and 25,000 images, this produced ~30-45 minute epochs — prohibitively slow for comparing 4 TL configurations. LPIPS was moved to a validation-only reporting metric (sampled on 10 batches), reducing epochs to ~6 minutes without losing the perceptual quality signal in the comparison.

Lesson: LPIPS is most valuable as a final training loss after hyperparameters are set, not as a per-batch gradient signal during architecture search.

9. Plan for Phase 5

9.1 Full Training with Best Config

- ▶ Train the final pSpResNet model for 20-30 epochs using the best HPO configuration (dropout_p=0.354, lr_backbone=5.4e-5, lr_head=1.3e-4)
- ▶ Use L1 + LPIPS (lambda_lips=1.461) as the training loss — LPIPS was only used for reporting in Phase 4 but should be in the gradient for Phase 5 to address blurriness
- ▶ Include VAE-augmented faces in training data to expand effective dataset size

9.2 Evaluation Experiments

- ▶ SSIM (Structural Similarity Index) on test set — Phase 3 and 4 only reported L1
- ▶ FID (Frechet Inception Distance) on 1,000 generated vs 1,000 real faces — standard GAN-era metric for synthesis quality
- ▶ LPIPS on full test set — not just sampled validation batches
- ▶ Per-identity consistency: given 5 different sketches of the same person, does the model produce consistent facial identity?

9.3 Ablation Studies

- ▶ Ablation 1: With vs without F.normalize — quantify the stabilisation contribution
- ▶ Ablation 2: With vs without VAE augmentation — measure F1/L1 difference on augmented training set
- ▶ Ablation 3: L1 only vs L1+LPIPS as training loss — quantify sharpness improvement from perceptual loss
- ▶ Ablation 4: Config C (best TL) vs Config A (frozen) at 20 epochs — does the gap widen or close with more training?

9.4 Error Analysis

- ▶ Identify failure modes: which sketch types produce worst reconstructions? (heavy hair occlusion, extreme pose, unusual lighting)
- ▶ Visualise attention / gradient maps to understand which edge regions the encoder focuses on
- ▶ Compare reconstruction quality by facial attribute (gender, age, expression) using CelebA attribute labels
- ▶ Plot reconstruction error distribution — are failures clustered or uniform across the dataset?

9.5 Report

- ▶ Full 16-20 page final report with all ablations, error analysis, and qualitative results
- ▶ At least 32 qualitative examples (4 rows × 8 columns: sketch, Phase 3, Phase 4, Phase 5, ground truth)
- ▶ Commit hash tagged as phase5-submission